# Unit 10 · Area, Surface Area and Volume Cheat Sheet

**Unit 10 focus:** area formulas, composite figures and units

## 1 AREA & UNITS

| | |
|---|---|
| **Area**: space inside a 2-D figure | **square units (cm², ft²)** |
| **Perimeter**: distance around | **linear units (cm, ft)** |
| **Surface area**: total area of the outside faces | **square units** |
| **Volume**: space inside a 3-D figure | **cubic units (cm³, ft³)** |

## 2 POLYGON AREA FORMULAS

| Figure | Area |
|---|---|
| rectangle | $A = lw$ |
| square | $A = s^2$ |
| triangle | $A = \frac{1}{2}bh$ |
| parallelogram | $A = bh$ |
| trapezoid | $A = \frac{1}{2}(b_1 + b_2)h$ |
| circle | $A = \pi r^2$ |

Height $h$ is always perpendicular to the base (not a slanted side).

## 3 TRAPEZOID EXAMPLE

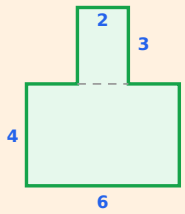
Average the two parallel bases, then multiply by the height.

**Example:** Bases 6 cm and 10 cm, height 4 cm
**Formula:** $\frac{1}{2}(6 + 10)(4) = \frac{1}{2}(16)(4)$
**Answer:** $32\ \text{cm}^2$

## 4 COMPOSITE FIGURES

Split into simple shapes; add (or subtract) their areas.

**Example:** A 6 × 4 rectangle with a 2 × 3 rectangle on top
**Areas:** $6 \times 4 = 24$ and $2 \times 3 = 6$
**Add:** $24 + 6$
**Answer: 30 square units**




## 5 CIRCLES

| Find | Formula | r = 3 |
|---|---|---|
| circumference | $C = 2\pi r$ | $6\pi \approx 18.8$ |
| area | $A = \pi r^2$ | $9\pi \approx 28.3$ |

## 6 QUICK REMINDERS & COMMON MISTAKES

**✓ Do remember**

- Square units for area, cubic units for volume.
- Use the perpendicular height.
- Keep $\pi$ unless asked to round.

**✗ Common mistakes**

- Slant side as height: use the perpendicular height
- Diameter for $r$: halve it
- Forgetting a face: list every face for surface area

# Unit 10 · Area, Surface Area and Volume Cheat Sheet

**More practice:** surface area and volume of prisms, cylinders, pyramids, cones and spheres

## 1 PRISMS

$V = Bh$ ($B$ = base area). $SA = Ph + 2B$ ($P$ = base perimeter).

**Example:** Rectangular prism 5 × 3 × 4
**Volume:** $5 \cdot 3 \cdot 4 = 60$
**SA:** $2(15 + 20 + 12) = 94$
**Answer:** $V = 60$ **units³,** $SA = 94$ **units²**

## 2 CYLINDERS

$V = \pi r^2 h$. $SA = 2\pi r^2 + 2\pi rh$ (two circles + the curved side).

**Example:** $r = 3$, $h = 8$
**Volume:** $\pi(9)(8) = 72\pi$
**SA:** $18\pi + 48\pi = 66\pi$
**Answer:** $72\pi$ **units³,** $66\pi$ **units²**

## 3 PYRAMIDS & CONES

| Solid | Volume | Surface area |
|---|---|---|
| pyramid | $\frac{1}{3}Bh$ | $B + \frac{1}{2}P\ell$ |
| cone | $\frac{1}{3}\pi r^2 h$ | $\pi r^2 + \pi r\ell$ |

$h$ = vertical height; $\ell$ = slant height (on the face). A pyramid holds $\frac{1}{3}$ of the matching prism.

## 4 SPHERES

$V = \frac{4}{3}\pi r^3$. $SA = 4\pi r^2$.

**Example:** $r = 3$
**Volume:** $\frac{4}{3}\pi(27) = 36\pi$
**SA:** $4\pi(9) = 36\pi$
**Answer:** $36\pi$ **units³ and** $36\pi$ **units²**

## 5 VOLUME EXAMPLE

Find the base area first, then multiply by the height.

**Example:** Cylinder $r = 4$, $h = 10$
**Base:** $\pi(4)^2 = 16\pi$
**Volume:** $16\pi(10)$
**Answer:** $160\pi \approx 502.7$ **units³**

## 6 CHECK YOURSELF

- Did I use the radius (not the diameter)?
- Vertical height for volume; slant height for surface area of pyramids and cones.
- Did I include every face?
- Are my units squared (area) or cubed (volume)?